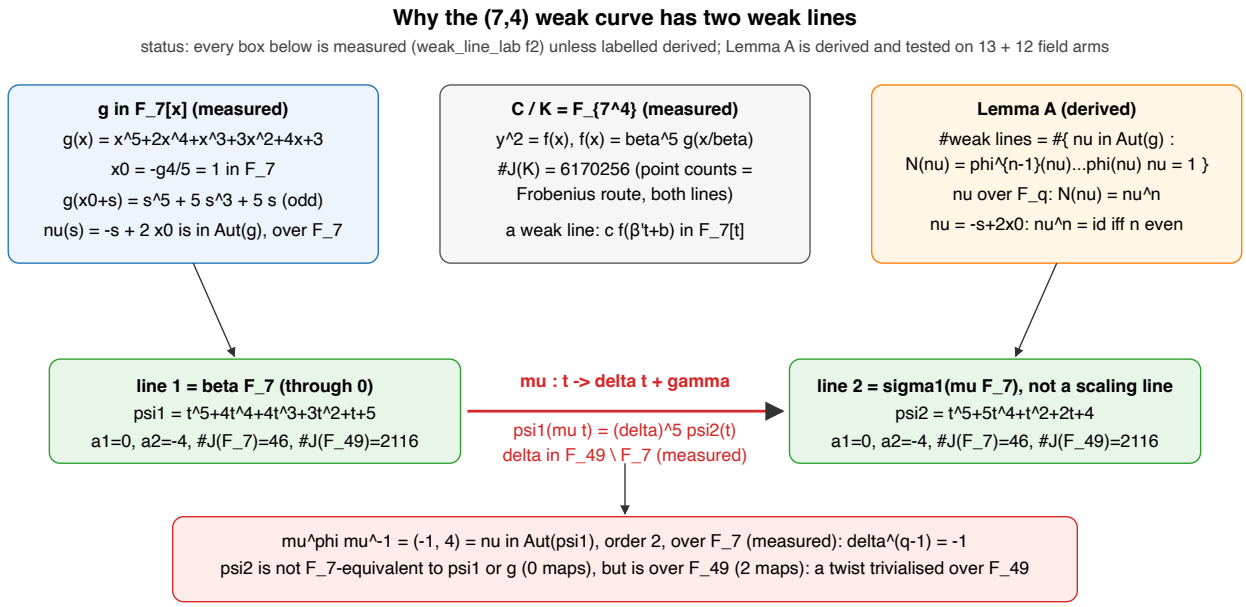
The second weak line at (7, 4): an affine symmetry of g, acting through F_{q²}

Date 2026-10-13. Protocol: `PROTOCOL.md`, frozen at commit `81dbf9136` before any run, no amendments. Code: `weak_line_lab.rs` (fork of `weak_curve_lab.rs`, same field/census code). Records: `results/`, environment in `results/run-20261013.env.txt`. Conductor: no control plane was reachable (`coord_start_work` returned 404), so no task id.

This answers the open item of the merged study `research/weak_subfield_hyperelliptic_decomposition_20261012/` (#1757, README E3: "Unexplained … second weak line at (7, 4)"). That study's frozen records are not edited. Structural diagnostic on toy fields; no speedup, sub-rho or deployed-curve claim.

Answer

The (7,4) weak curve is $f(x) = \beta^5\, g(x/\beta)$ with $g \in F_7[x]$ odd about an F_7-point: $g(x_0 + s) = s^5 + 5s^3 + 5s$ with $x_0 = 1$ (measured). So $\nu(s) = -s + 2x_0$ is an affine symmetry of g defined over F_7, and because $n = 4$ is even, $N(\nu) = \nu^4 = \mathrm{id}$, which by Lemma A (derived below) gives exactly one extra weak line: $\sigma_1(\mu F_7)$ with $\mu(t) = \delta t + \gamma$, $\delta \in F_{49} \setminus F_7$, $\delta^{q-1} = -1$. Both H1 (affine symmetry) and H2 (quadratic subfield) hold, as one mechanism: the second F_7-form is the twist of the first by the class of ν, trivialised over F_{49}. H3 (coincidence with trivial $\mathrm{Aut}(g)$) is falsified: $\mathrm{Aut}(g)$ has order 2 (the full scan over all 2400 scalars).



Derivation (Lemma A; derived, then tested)

Let $\psi_1 \in F_q[t]$ be monic quintic and $f = \psi_1 \circ \sigma_1^{-1}$ with σ_1 affine (the known line; $\psi_1 = g$ for the E3 curve). For $\mu \in \mathrm{Aff}(K)$, the line $\sigma_1(\mu F_q)$ is weak iff $\psi_1(\mu t)$ is, up to a scalar, in $F_q[t]$. Both $\psi_1(\mu t)/\zeta_\mu^5$ and its φ-conjugate $\psi_1(\mu^\varphi t)/\zeta_\mu^{\{5q\}}$ are monic, so weakness is equivalent to $\psi_1(\mu\, t) = \psi_1(\mu^\varphi\, t)\cdot(\mathrm{scalar})$, i.e. $\nu := \mu^\varphi\, \mu^{-1} \in \mathrm{Aut}(\psi_1) = \{(\zeta, c) : \psi_1(\zeta s + c) = \zeta^5 \psi_1(s)\}$. A line determines μ up to right multiplication by $\mathrm{Aff}(F_q)$, which does not change ν. Iterating $\mu^\varphi = \nu\mu$ n times gives the cocycle condition $N(\nu) = \varphi^{n-1}(\nu)\circ\ldots\circ\varphi(\nu)\circ\nu = \mathrm{id}$, and conversely every such ν is a coboundary by Hilbert 90 for G_a and G_m (so for $\mathrm{Aff} = G_a \rtimes G_m$). Therefore

$$\#\text{weak lines} = \#\{\ \nu \in \mathrm{Aut}(g) : N(\nu) = \mathrm{id}\ \}. \qquad \text{(Lemma A)}$$

Corollaries (derived): (i) there is always the line $\nu = 1$; (ii) a second line needs $\mathrm{Aut}(g) \neq 1$; (iii) a nontrivial ν fixes ∞ and permutes the five roots, so has order 2, 4 (fixing a root) or 5; (iv) g odd about $x_0 \in F_q$ gives ν over F_q, $N(\nu) = \nu^n$, extra line iff n even, with $\delta \in F_{q^2}$. The parity step (iv) is checked in Lean: `lean/InvolutionNorm.lean` proves $(-1,c)\circ(-1,c) = \mathrm{id}$ and $\nu^n = \mathrm{id} \leftrightarrow n$ even for affine maps over `Int` (`lake build`, `lean/build.log`; the theorems use the standard axioms `propext`, `Classical.choice`, `Quot.sound` via `omega` and `simp`; no Mathlib). Lemma A itself (Hilbert 90 step) is not formalised.

Results

F1 reproduction (measured)

`./weak_line_lab f1` prints the E3 weak-curve record at (7,4); `cmp` against the line recorded in `run-20261013T0404Z.jsonl` is silent: byte-identical (`results/f1.jsonl`, `results/f1.recorded.jsonl`).

F2 the two lines (measured; `results/f2.jsonl`)

	line 1	line 2
line	βF_7, contains 0 (scaling line)	not through 0
monic ψ on the line	$t^5+4t^4+4t^3+3t^2+t+5$	$t^5+5t^4+t^2+2t+4$
Frobenius `(a1, a2)`	(0, −4)	(0, −4)
`#J` over F_7, F_{49}	46 = 2·23, 2116	46, 2116
`#J(K)` by Frobenius vs point counts	6170256, agrees	6170256, agrees

- $g = x^5+2x^4+x^3+3x^2+4x+3$ over F_7; centred form s^5+5s^3+5s, so `Aut(g) = {id, s ↦ −s+2}` (full scan over all 2400 ζ; the restricted scan over roots of unity agrees), both with norm 1: Lemma-A count 2 = census 2.
- The K-affine map between the lines has scalar $\delta \in F_{49} \setminus F_7$; $\nu = \mu\varphi\mu^{-1} = (-1, 4)$ lies in $\mathrm{Aut}(\psi 1)$, order 2, defined over F_7.
- $\psi 2$ is **not** F_7-equivalent to $\psi 1$ or g (0 affine maps over F_7), but there are 2 over F_{49} (the two differ by ν) and 2 over K: the second F_7-form is a twist trivialised over F_{49}, not the same form. Both forms have the same Frobenius data $(0, -4)$; with $a_1 = 0$ the quadratic twist already has the same L-polynomial, so equal L-polynomials do not distinguish them here.
- Line 2 is not the image of line 1 under `Aut` of f: ν maps βF_7 to itself ($c \in F_7$), which is why the extra line is not an orbit effect.

F3 control census (measured; `results/f3.jsonl`)

300 fresh weak curves per field (2000 at (7,4)), same sampler as E3. Lemma-A count = census count on every one of the 5600 curves (12 fields at 300, (7,4) at 2000): **0 mismatches in 13 of 13 fields**; shortcut census = exhaustive census on all checked curves (all where $|K| \le 2401$, 5 at (11,4) and (13,4)); restricted vs full `Aut` scan agree wherever both were run.

(q,n)	curves	weak-line counts (curves)	with g odd about an F_q-point
(3,2)	300	1: 274, 2: 26	26 (all 2 lines)
(7,2)	300	1: 297, 2: 2, 4: 1	3
(11,2)	300	1: 299, 4: 1	1
(13,2)	300	1: 299, 2: 1	1
(3,4)	300	1: 243, 2: 27, 5: 30	27
(7,4)	2000	1: 1966, 2: 21, 4: 5, 5: 8	26 (21 + 5)
(11,4)	300	1: 294, 2: 5, 4: 1	6
(13,4)	300	1: 299, 2: 1	1
(3,6)	300	1: 271, 2: 29	29
(3,3) (7,3) (11,3) (3,5)	300 each	all 1	27, 3, 2, 18, none with a second line

So an extra weak line is not special to this g but is not generic either: at (7,4) it occurs in 34 of 2000 fresh weak curves (1.7%), always with $\mathrm{Aut}(g) \neq 1$. Odd n never gives one for an odd g ($\nu^n \neq \mathrm{id}$), matching the E3 data (every odd-n field had one line). The (3,4) and (7,4) rows include curves with 5 lines: order-5 symmetry $\zeta_5 \in F_{q^4}$, norm 1 (this is the E3b x^5+c situation). Note the (11,4) sample has 6 curves with an odd g against 2.2 expected from the exact rate below (one field of 13; I did not correct for that; the Lemma-A/census agreement is exact on those curves).

F4 exact population by g (measured tally of Lemma A; `results/f4.jsonl`)

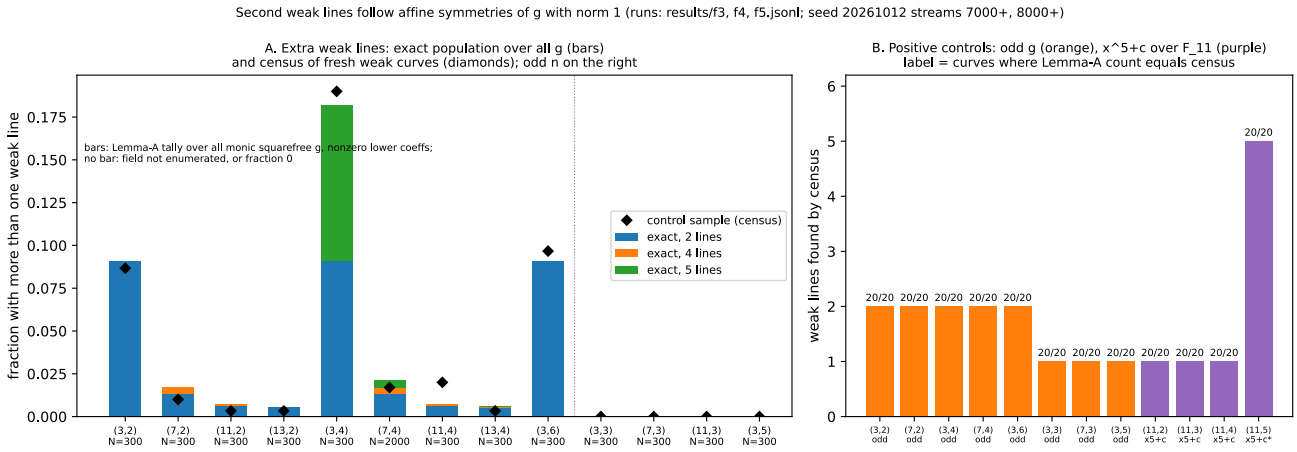
Every monic squarefree $g \in F_q[x]$ with nonzero lower coefficients (ζ restricted to roots of unity of order 2, 4, 5):

(q,n)	g	1 line	2 lines	4 lines	5 lines
(7,2)	6798	6684	90	24	0
(7,3)	6798	6798	0	0	0
(7,4)	6798	6654	90	24	30
(11,4)	91690	91010	600	80	0
(13,4)	231036	229656	1128	120	132

At (7,4), 144 of 6798 g (2.1%) give more than one weak line; the E3 curve is one of the 90 two-line g (odd g, an F_7-rational involution). Fields (3,·), (11,·), (13,·) are in the JSONL. Under the E3 sampler at the three even-n fields it ran (not (5,4): char 5, below), the chance of at least one extra line was therefore roughly 20% (dominated by (3,4)); the E3 observation needed no special explanation beyond this symmetry.

F5 positive controls (measured; `results/f5.jsonl`)

20 constructed curves per arm; exhaustive census where $|K| \le 2401$, else shortcut. Odd g: 2 lines at (3,2) (7,2) (3,4) (7,4) (3,6) and 1 line at (3,3) (7,3) (3,5), all 20/20 as Lemma A predicts. x^5+c over F_{11}: 1 line at n = 2, 3, 4 ($\zeta_5 \in F_{11}$, $N = v^n \neq id$), 20/20; supplementary arm added after the freeze (labelled * in the graph): 5 lines at (11,5), 20/20, matching E3b's (11,5) row of 5 lines.



Graph source: `results/f3.jsonl`, `f4.jsonl`, `f5.jsonl` via `reports/plot.py` (layout only). Panel A: exact population fractions (bars) and control census fractions (diamonds), seed `20261012` streams `7000+`; panel B: positive controls.

Scope, limits, what stays open

- Lemma A is derived here and supported by 0 mismatches in 25 field arms (13 control fields + 12 positive-control arms, 20 curves each) and the exact population tallies; it is not formalised beyond the parity step.
- Char 5 is outside Lemma A (translations are symmetries there): the E3 (5,4) weak curve with 1 line is not covered by this study.
- E3b's x^5+c counts (5 at (3,4) and (7,4)) and x^5+cx count (4 at (7,4)) are consistent with the lemma by hand (order-5 and order-4 symmetries with norm 1), but E3b counts only `square_ok` lines, so I did not compare the other rows and did not rerun them.
- The relation spans `H` of the second line were not computed; whether the second form changes the collapse bound is untested here. Both forms have `#J(F_7) = 46`, `#J(F_49) = 2116`, so the P1 bound is the same size.
- Nothing here concerns deployed curves or any speedup.

Graphs checked and left unchanged

No canonical graph changes: this follow-up adds no ECDLP rung, no timing (so no `benchmarks/` entry); `docs/index-calculus-scoreboard.html`, `docs/ic/progress-timeline.json` and `docs/ic/boundary_targets.json` are unaffected.

Reproduce

```
rustc --edition 2021 -O weak_line_lab.rs
./weak_line_lab f1 | cmp - results/f1.recorded.jsonl
./weak_line_lab f2; ./weak_line_lab f3; ./weak_line_lab f4; ./weak_line_lab f5
python3 reports/plot.py                                    # layout only
typst compile --root . REPORT.typ REPORT.pdf
cd lean && lake build
```